

1 Single Neuron/Perceptron

$$\sigma\left(\sum_{i=1}^n X_i W_i + b\right) \quad (1)$$

2 Multi layer Perceptron

2.1 Notations

W_{jk}^l is weight from k index in row $l - 1 \rightarrow j$ index in l in layer l
Lets say we have the below matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

then it is a matrix with j in current layer rows and k in previous layer columns
and W_{jk} will be a node to node weight.

b_j^l is bias for layer l at index j in layer l .